

Subject Description Form

Subject Code	ME548
Subject Title	Computer Aided Product Analysis
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Students should have basic knowledge of Mechanical Engineering, Building Service Engineering, Civil and Structural Engineering, Manufacturing Engineering, Product Design and Development, and Materials Science.
Objectives	To provide students with CAD/CAE knowledge and a good understanding of CAD and CAE technologies. The subject covers CAD, Finite Element Method, EM, computer-aided analysis, integration of CAD and CAE, and virtual engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - possess knowledge in the area of principle and formulations of finite element method, computer-aided design and engineering; - analyze static and dynamic stress and strain behaviours and thermal engineering of structures and products using CAD and CAE techniques; - apply their knowledge and skills to design and develop products; and - have recognition of the need for, and an ability to engage in life-long learning.
Subject Synopsis/ Indicative Syllabus	1. Computer-Aided Design (CAD) 1.1 Introduction to computer-aided design. 1.2 Product geometric models: Wireframe model, Surface model, Solid Model. 1.3 CAD geometry modelling: Wireframe geometry modelling, Surface modelling, Solid modelling. 1.4 CAD application: CAD-enabled injection mould design. 2. Computer-Aided Engineering (CAE) 2.1 Introduction to Finite Element Method (FEM) and FEM analysis 2.2 FEM Principles, Methods to establish FEM formulation, shape functions, element type and selection, FEM mesh generation. 2.3 Finite Element Modeling and Simulation: Finite element modelling and simulation, stress computation and analysis, strain computation and analysis, and thermal engineering analysis. 2.4 Finite element simulation enabled solution generation. 3: CAD/CAE applications 3.1 Data exchange standards. 3.2 CAD and CAE integration. 3.3 CAD/CAE enabled deformed product design and development. 3.4 CAD/CAE enabled cast product design and development. 4. Simulation-enabled design optimization: 4.1 Geometric size design and optimization. 4.2 Geometric shape design and optimization.

Teaching/Learning Methodology	1. The teaching and learning methods include lecture/tutorial sessions, homework assignments, tests, case study reports and examinations. 2. The continuous assessment and examination are aimed at providing students with the integrated knowledge required for computer-aided analysis. 3. Technical/practical examples and problems are raised and discussed in class/tutorial sessions. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework assignment</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Case study report and presentation</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes				a	b	c	d	1. Lecture	√	√	√	√	2. Tutorial	√	√	√	√	3. Homework assignment	√	√	√	√	4. Case study report and presentation	√	√	√	√											
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Reading List and References	1. Fuh Y., Zhang Y.F., Nee A.Y.C., Fu M.W., Computer-aided injection mould design and manufacture, Marcel Dekker, Inc., New York, April 2004 2. Law A. M. and Kelton D. W., Simulation Modeling and Analysis, McGraw-Hill, latest edition. 3. Przemieniecki, J. S., Finite Element Structural Analysis, New Concepts, AIAA, latest edition. 4. Fu M.W., Design and Development of Metal Forming Processes and Products aided by Finite Element Simulation, Springer International Publishing AG, 2016.
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